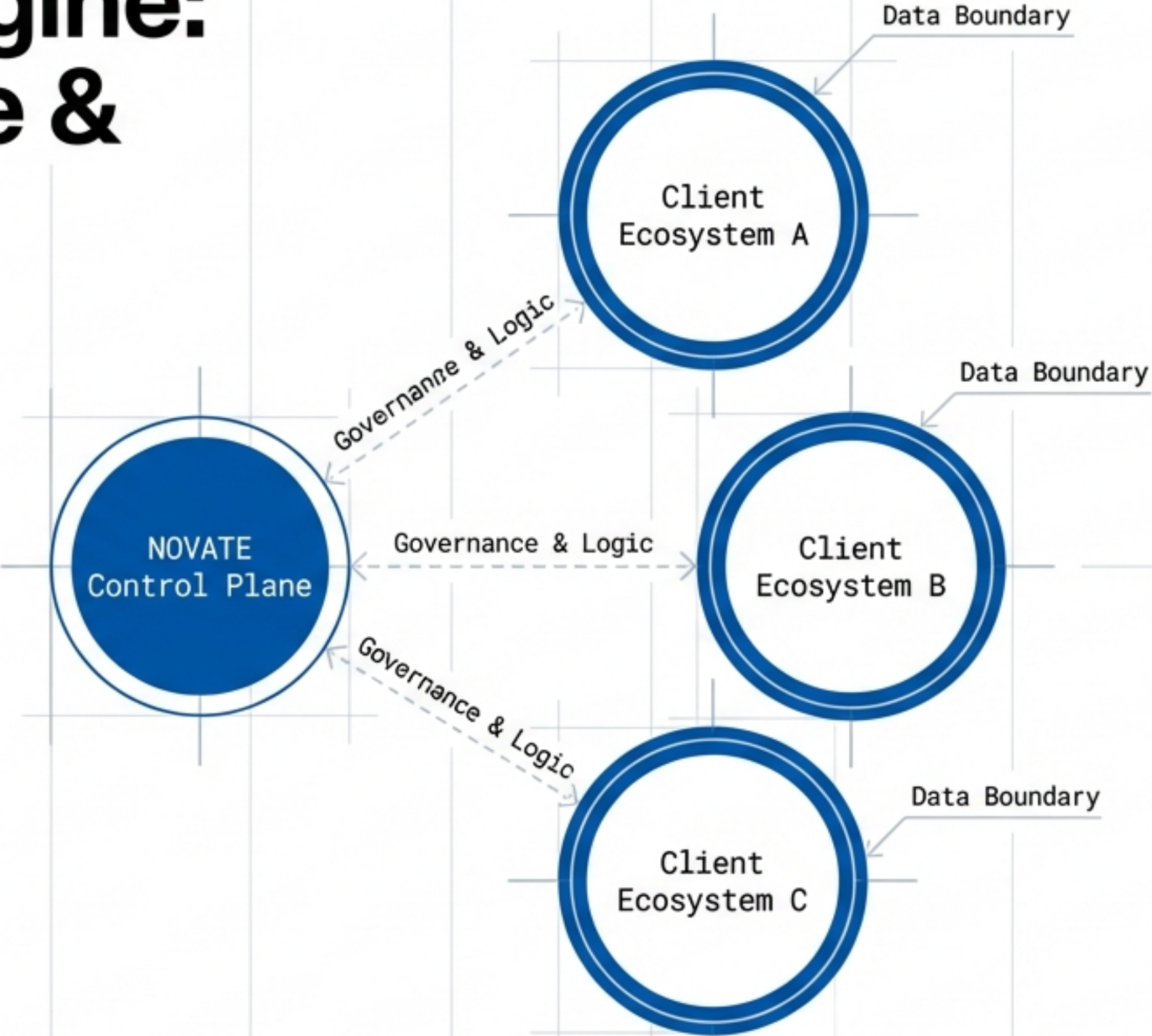


Sentra's Innovation Engine: Reference Architecture & Operational Strategy

A Multi-Tenant Control Plane
for Defensible Innovation



OPERATOR MODEL
NOVATE runs the engine;
clients own the outcomes.

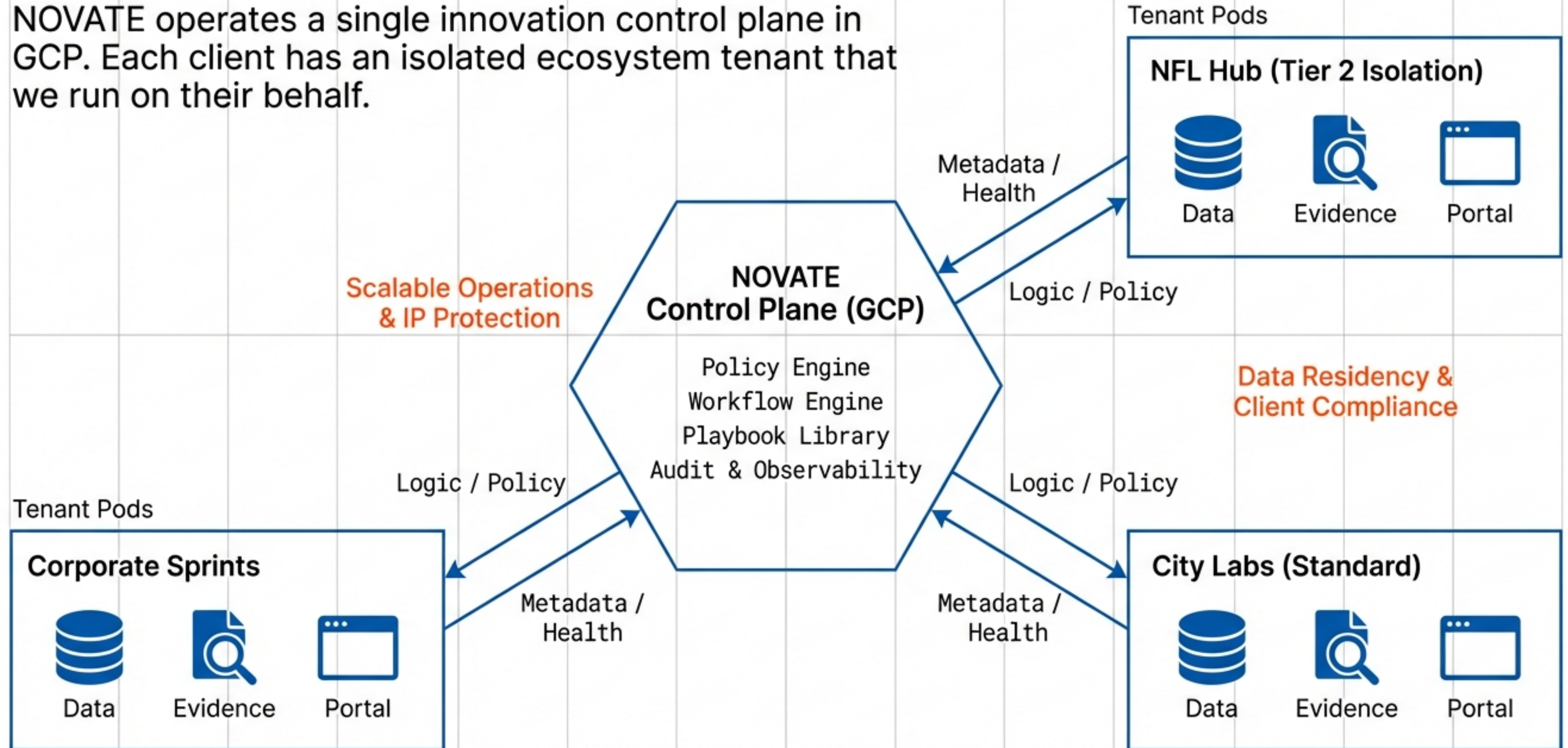
ISOLATION
Enterprise-grade tenant data
boundaries.

INTEGRATION
Connects to multi-cloud
client environments.

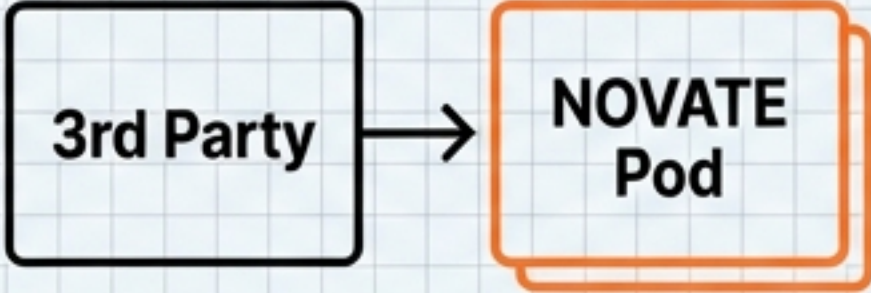
STANDARDIZATION
Powered by NIN (NOVATE
Innovation Network) standards.

One Control Plane. Many Ecosystems.

NOVATE operates a single innovation control plane in GCP. Each client has an isolated ecosystem tenant that we run on their behalf.



A Single Engine Supporting Three Innovation Modes

 Mode 1: NOVATE-Hosted Community / Research Focus	 Mode 2: Managed Innovation Client Programs	 Mode 3: White-Label Enterprise NFL Tier / Critical R&D
		
• Tenant: NOVATE Foundation • Data: Stored in NOVATE GCP • Portal: Native Sentrais Participant Portal • Use Case: Public hackathons, open innovation	• Tenant: Client-specific isolated pod • Intake: Native or 3rd Party (Submittable/Typeform) • Governance: System-of-record is Sentrais • Use Case: Corporate Challenges	• Tenant: Hard Isolation (Tier 2 / Dedicated DB) • Controls: Strict COI, IDS gating, Audit trails • IP: NOVATE Playbooks deployed as templates • Use Case: Internal R&D, IP Generation

The Architectural Stack: Core, Tenant, and Integration

Layer 1: NOVATE Control Plane (The Operator)

Context: Hosted in NOVATE GCP



Policy Engine
(Permissions/Gating)



Workflow Engine
(Stage Gates)



Registries
(Playbooks/Rubrics)



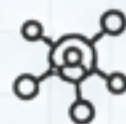
Observability & Governance Ops

Layer 2: Client Ecosystem Pods (The Tenant)

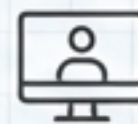
Context: Tenant-Isolated Resources



Tenant Data Boundary
(Encryption Keys/DB)



Ecosystem Objects
(Challenges/Submissions)



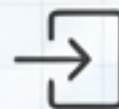
Client Portals
(Exec/Reviewer Views)



Branding Overlays

Layer 3: Channel & Integration Layer

Inter: Adapters & Connectors



Intake Adapters
(Native / 3rd Party)



Client Systems
(SSO, File Stores)

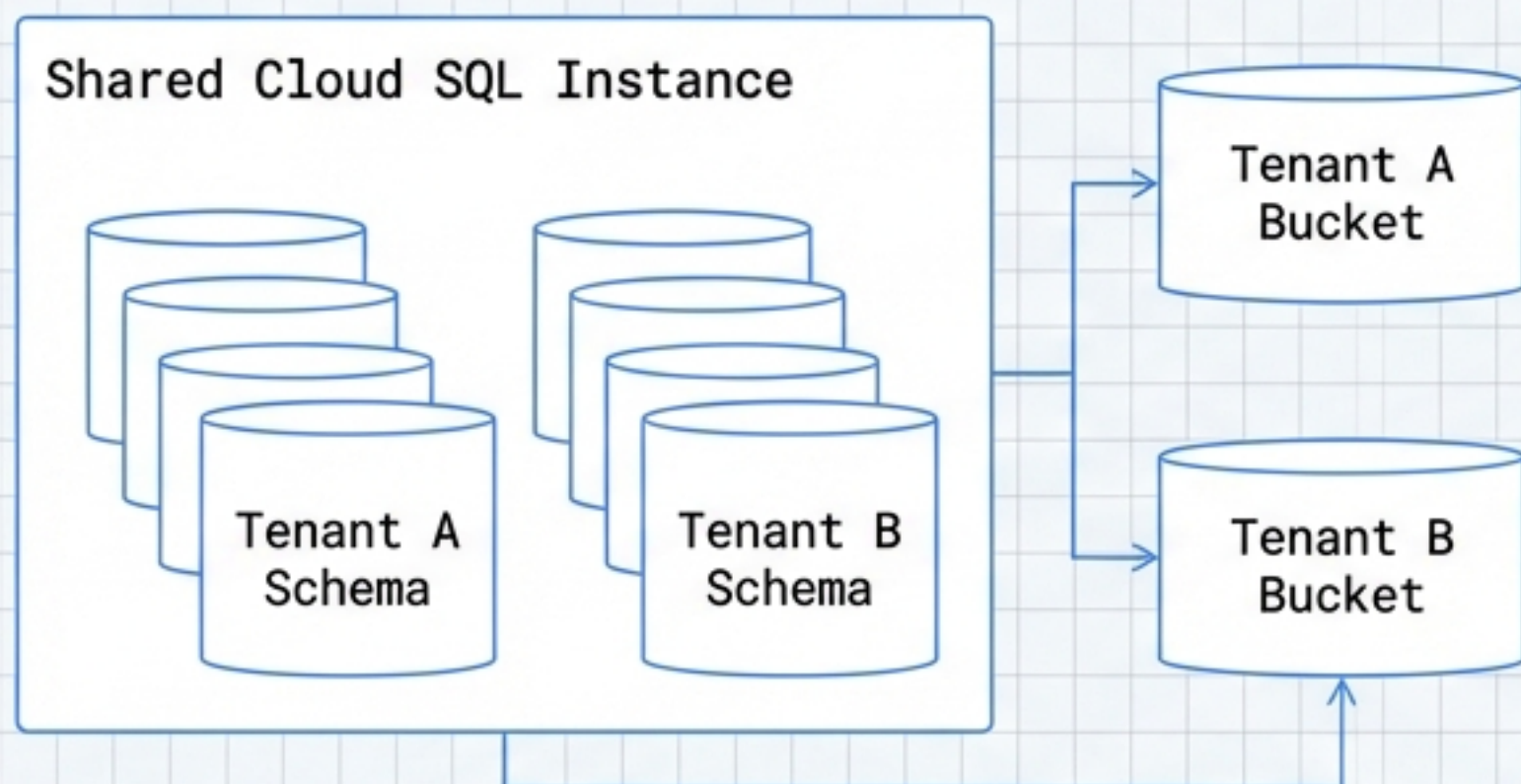


Data Export Connectors

Tenant Isolation & Data Security Strategy

Control plane is shared; data plane is tenant-isolated.

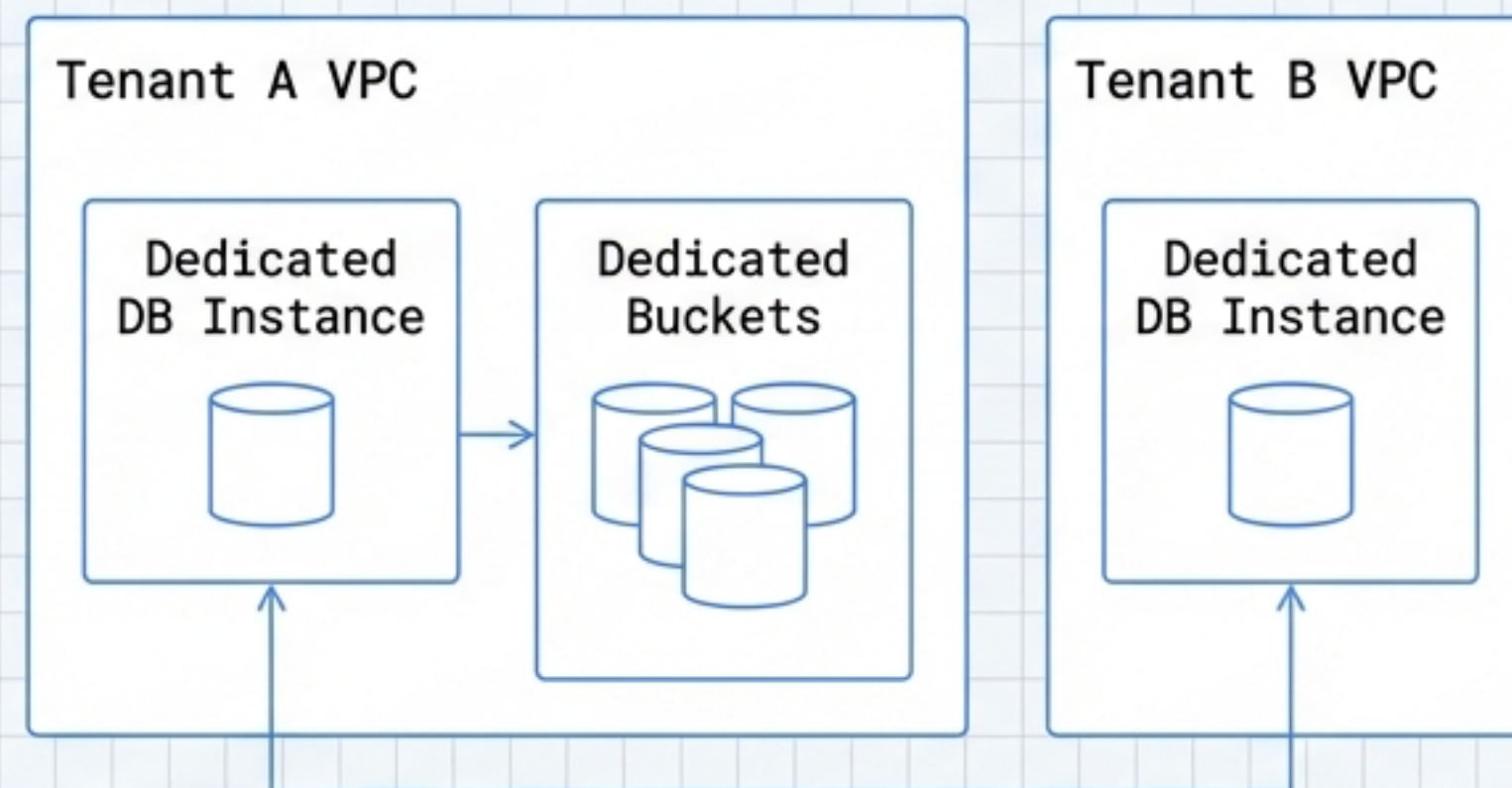
Tier 1 (Standard Isolation)



Key Features:

- Schema-per-tenant isolation
- Shared Compute Resources
- CMEK Support

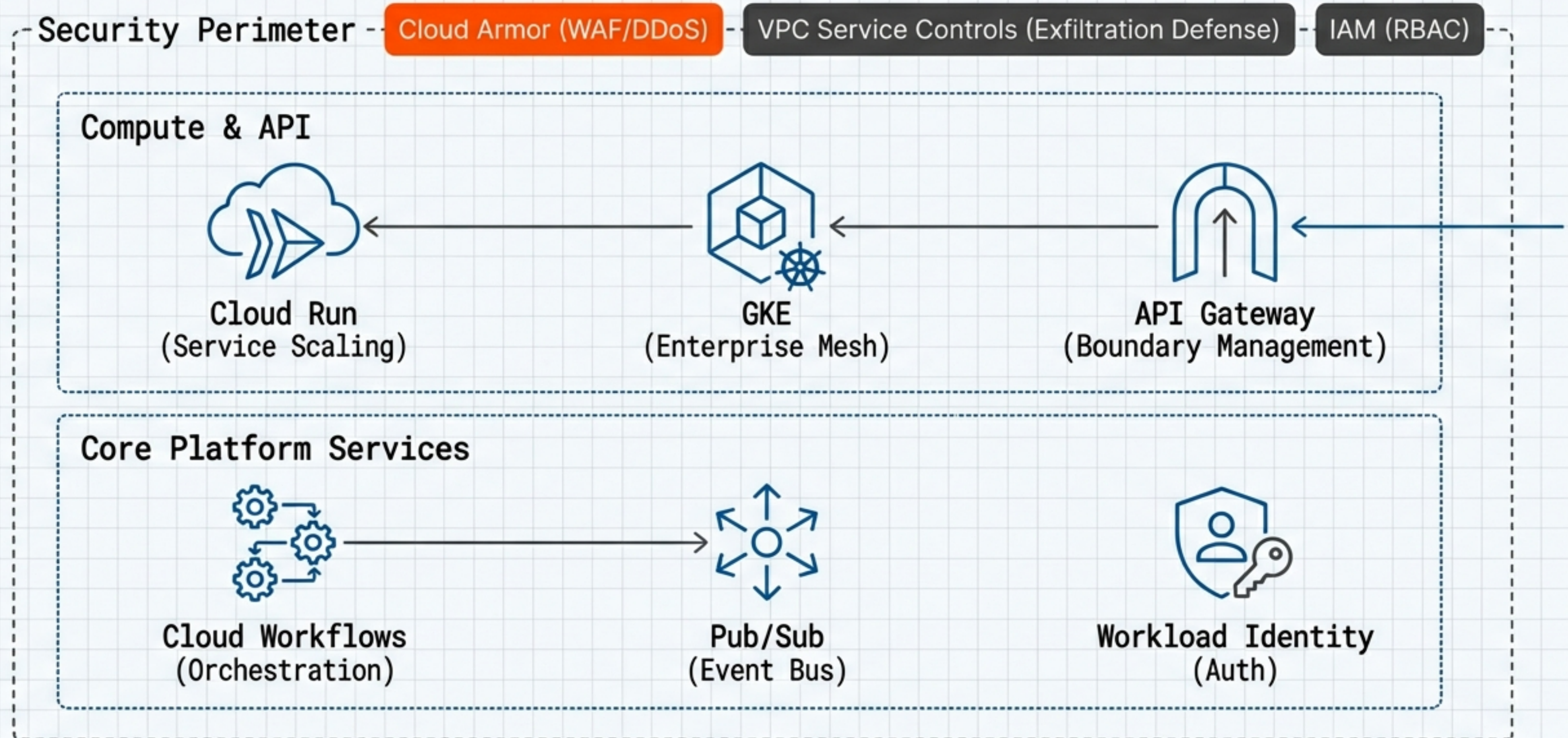
Tier 2 (NFL / Critical Isolation)



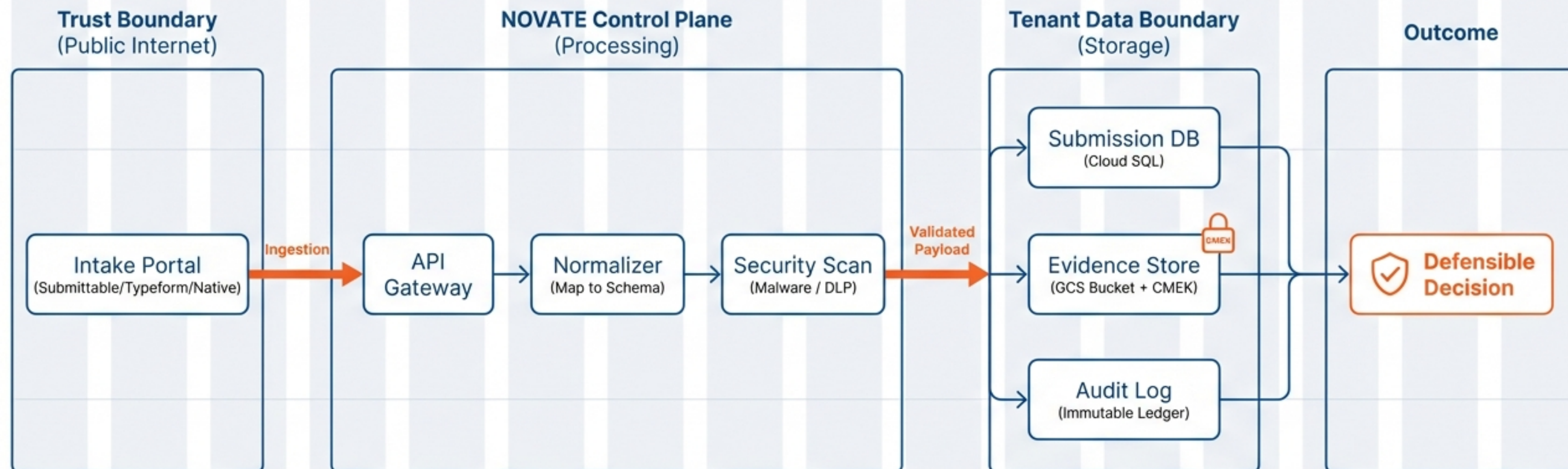
Key Features:

- Dedicated Database Instances
- Separate VPCs / Projects
- Strict Network Segmentation

GCP Reference Stack & Service Topology



Data Flow Architecture: From Intake to Defensibility



Security Note: Third-party intake is merely a channel. Sentrais is the system of record. All external data crosses a trust boundary before storage.

Multi-Cloud Integration Strategy

Engine in GCP, Integrate Everywhere.

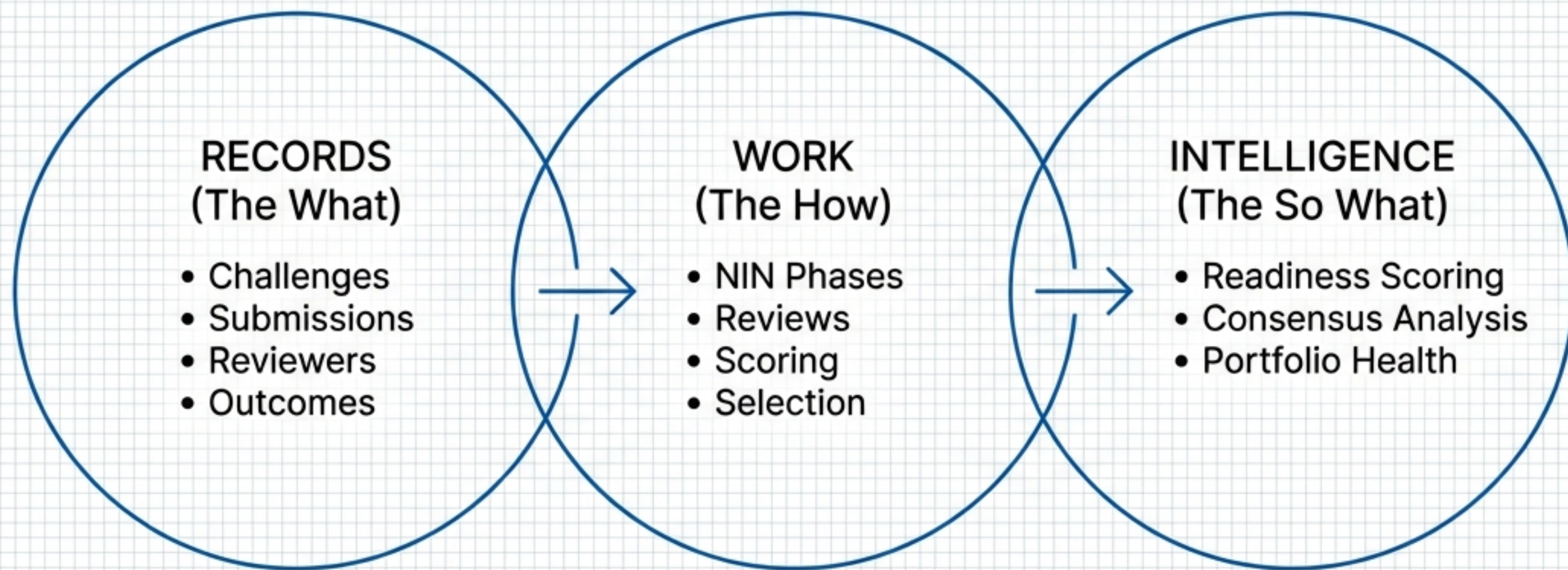


Philosophy: We **do not move the engine**; we move the data via connectors. This solves the **"Shadow IT" problem** by centralizing governance while preserving tool flexibility.

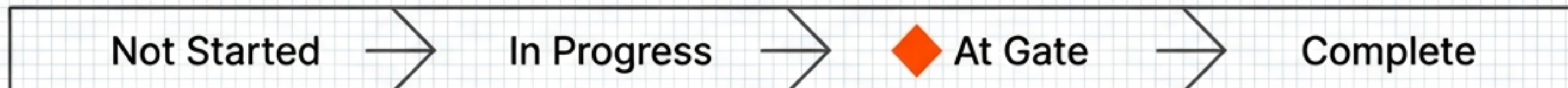
Innovation Orchestration Standards (NIN)

Innovation feels disciplined, not experimental.

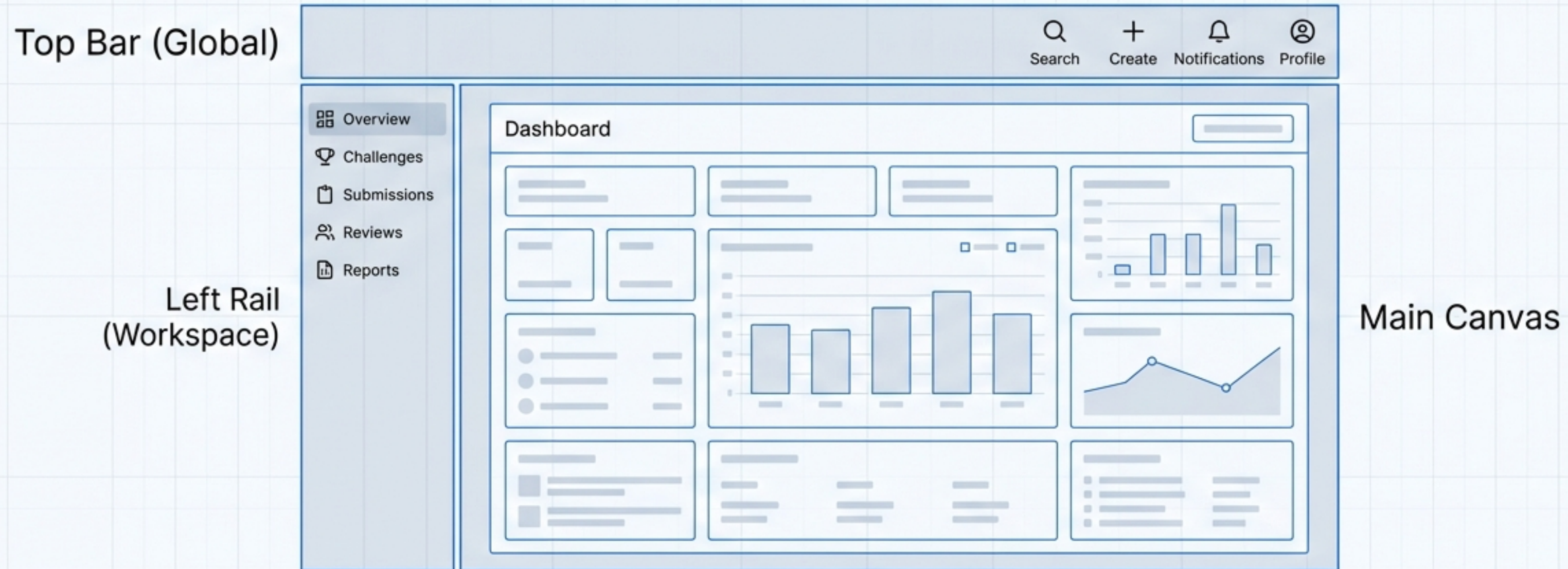
Problem-to-Outcome Pattern



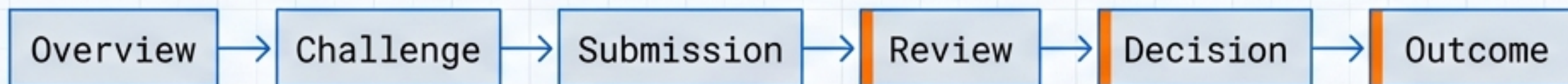
Workflow
Primitives



The Design System: Layout & Navigation



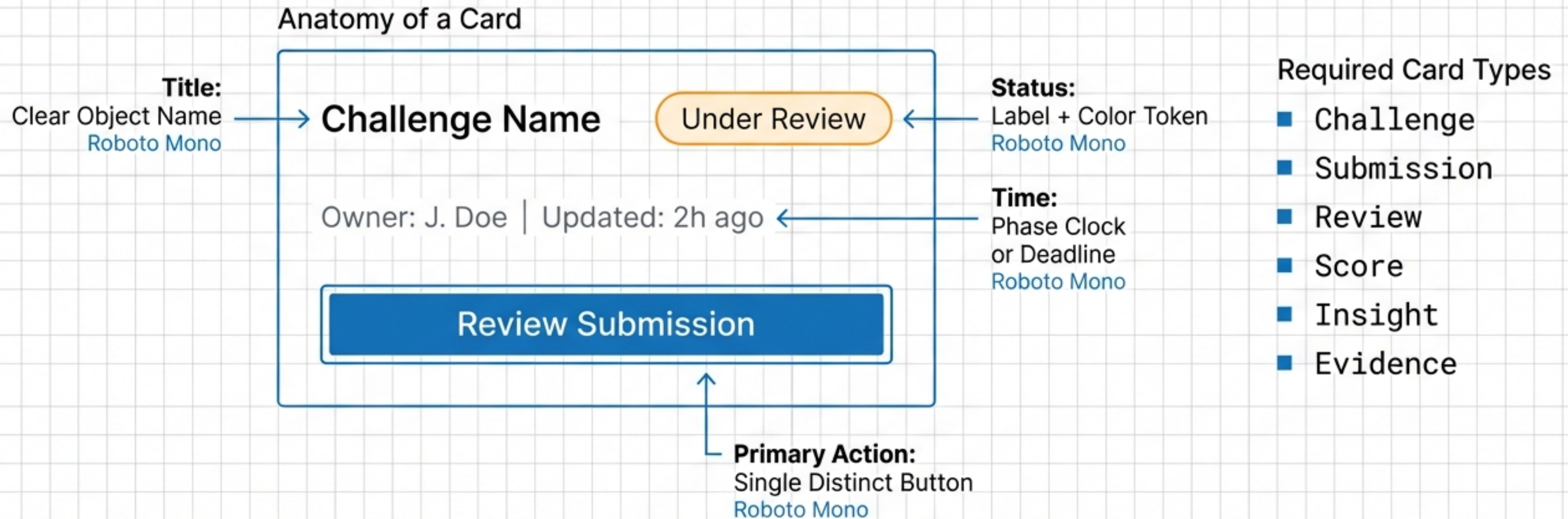
The Canonical Journey



Philosophy: Calm in complexity.
Uses standard design tokens to ensure consistency with Operations.

The Card System: Scannability & Action

Understandable and actionable in under 5 seconds.



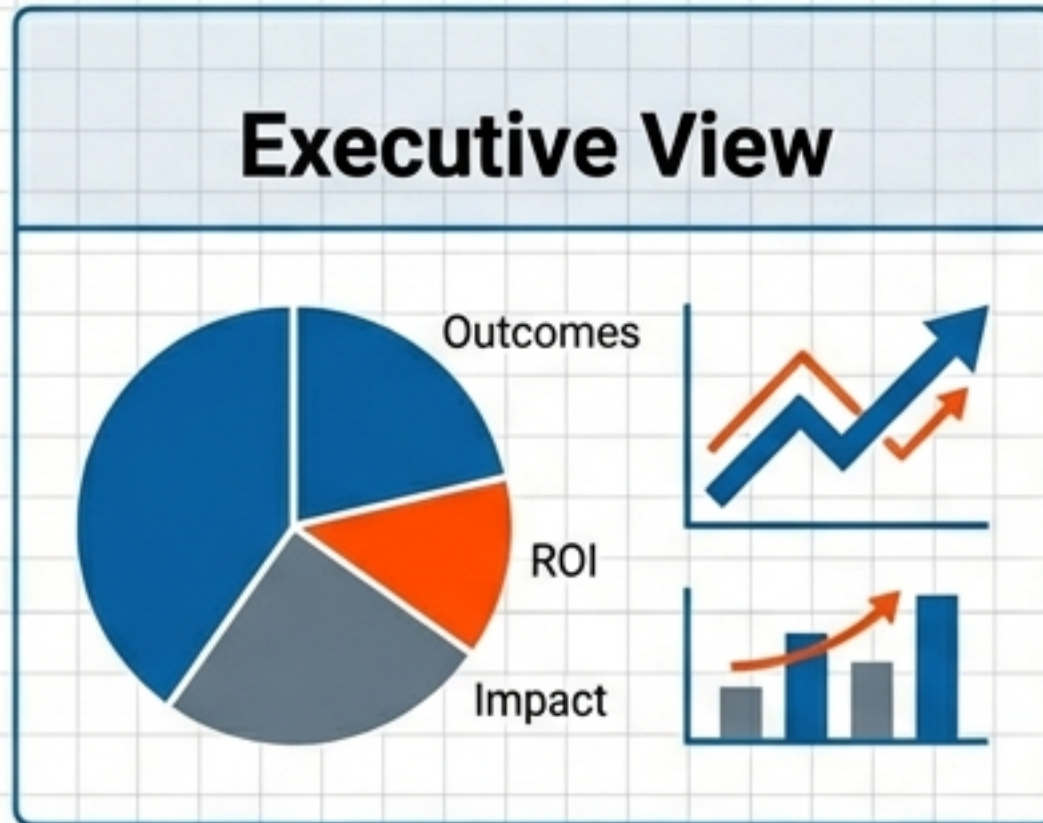
Governance & Permissions Model

ROLES	ACCESS LEVELS			
	Read / Submit	Score / Comment	Manage Gates	Evidence / Logs
	Observer	✓		
	Reviewer	✓	✓	
	Operator	✓	✓	
	Auditor	✓		✓

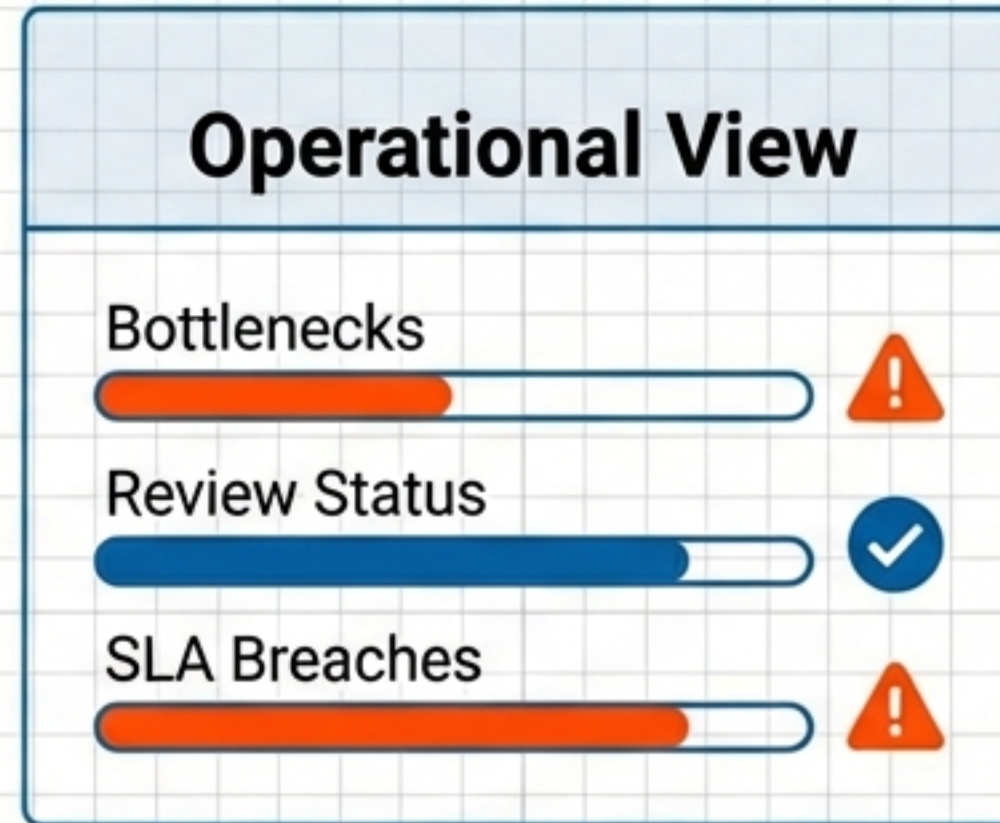
Context Awareness:
Permissions change dynamically based on Phase (early vs. late) and Sensitivity (blind reviews).

Security Principle:
Least privilege by default. All overrides generate audit log entries.

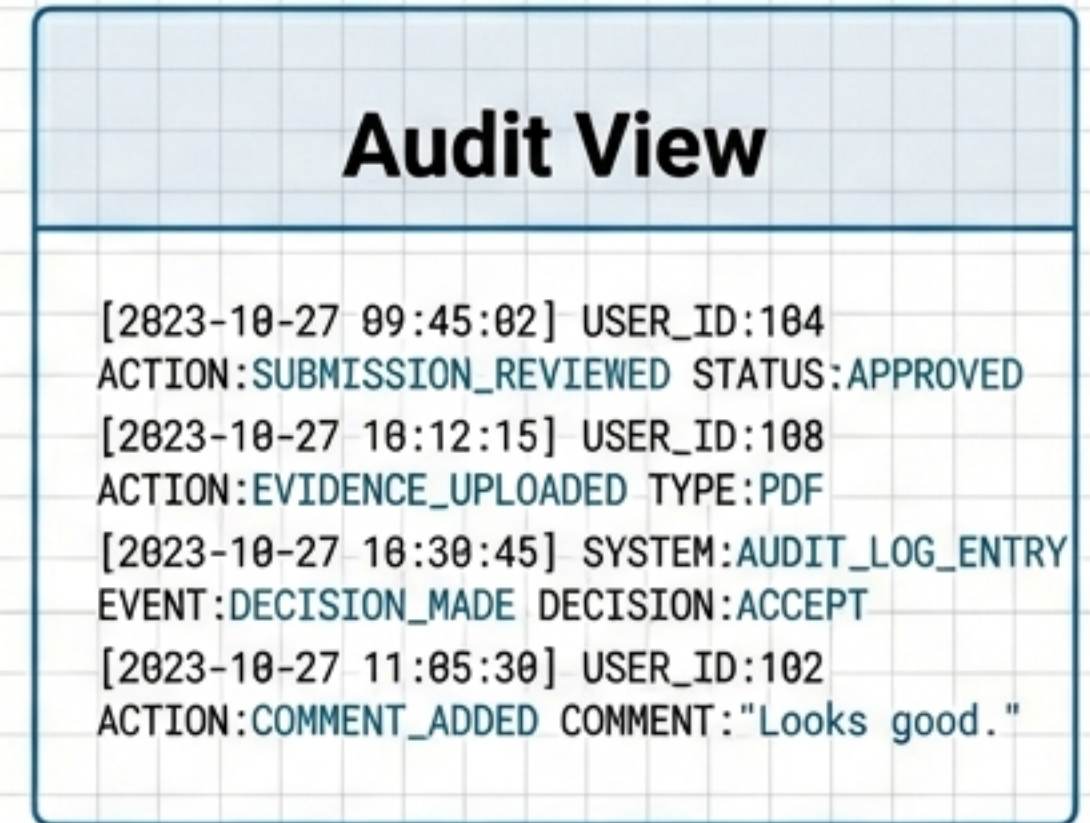
Reporting, Intelligence & Defensibility



Portfolio Health
Outcomes, ROI, Impact



SLA & Queues
Bottlenecks, Review Status



Evidence Ledger
Completeness, Decision History

Insight Governance: Insights must show: **Observation** + **Impact** + **Suggested Action**.
Innovation intelligence supports decisions; it does not replace them.

Why This Architecture Wins

Scalable operations for NOVATE. Defensible independence for Clients.

Value for CLIENT

Brand Independence: Their logo, their terminology.



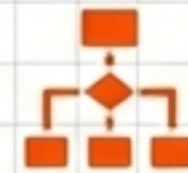
Rigorous Compliance: NFL-grade security and auditability.



Flexible Integration: Fits into existing multi-cloud worlds.



Value for NOVATE



Operational Scale: Manage many ecosystems from one console.



IP Protection: Playbooks remain central and protected.



Central Visibility: Portfolio insights without crossing data boundaries.

The Result: Innovation that is standardized, secure, and scalable.

Appendix: Glossary & Key Concepts

Control Plane	The multi-tenant, NOVATE-operated layer containing logic, policy, and workflow.
Tenant Pod	The isolated, client-specific environment containing data, encryption keys, and evidence.
CMEK	Customer-Managed Encryption Keys (client holds the keys to their data).
NIN	NOVATE Innovation Network standards (the workflow methodology).
Playbook	NOVATE IP providing the template logic for innovation challenges.

Core Definition: NOVATE operates a single innovation control plane in GCP. Each client has an isolated ecosystem tenant that we run on their behalf.